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RSC1 - BUSINESS INTELLIGENCE & DATA MINING LAB

25060 & 25062

Exploring Patterns in Real World Data:

Insights Through Visualization

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TTh 14:00 - 17:00



I. Introduction

Background / Introduction

Social media has become an essential part of daily life, influencing communication, entertainment, education, and social interaction across all age groups. While digital platforms provide convenience and connectivity, excessive and uncontrolled use has led to emerging concerns related to mental health, sleep deprivation, and behavioral addiction. Social media addiction is now recognized as a growing public health issue, particularly among adolescents and young adults who are most exposed to highly engaging online content.

As digital habits become more deeply integrated into everyday life, understanding *how* and *why* individuals develop addictive usage patterns requires more than simple observation. The field increasingly benefits from data mining, which enables researchers to examine large datasets and uncover hidden relationships between demographic factors, behavioral habits, and health outcomes. In the context of social media addiction, data mining allows analysts to explore questions such as:

- Which age groups are most vulnerable to high addiction levels?
- How does average daily use relate to sleep duration or mental health?
- Which platforms are most associated with excessive usage?
- Do factors like gender, device type, or screen time awareness influence addiction severity?

Using structured techniques such as classification models, clustering algorithms, and correlation analysis, data mining can detect patterns that are not immediately visible through traditional methods. In this study, approaches like Decision Trees, Random Forest, and K-Means Clustering were applied to identify predictors of addiction levels and to group individuals based on shared behavioral and demographic characteristics. These techniques provide deeper insights into how digital behavior differs across teens, adults, and seniors, and how these behaviors impact mental health and sleep.

The field also greatly benefits from data visualization, which translates complex data patterns into intuitive and easy-to-understand graphics. Visual tools such as bar



charts, cluster plots, and comparative dashboards present the relationships between usage habits and well-being in a form that is accessible to both technical and non-technical audiences. By visualizing trends, such as the link between high usage and lower sleep hours, or the platforms associated with high addiction levels, findings become clearer, more meaningful, and easier to interpret.

Overall, studying social media addiction through data mining and visual analytics creates a stronger, evidence-based understanding of how digital habits shape human behavior. The insights gained from these processes can support awareness campaigns, guide healthier technology use, and inform educational or policy interventions aimed at reducing the negative impacts of excessive social media consumption.

Dataset Description

- **Dataset name/source**

The study utilizes two real-world datasets related to social media behavior and addiction, both obtained from reputable open-data platforms. The first dataset, titled Social Media Addiction Dataset, was sourced from Kaggle, a platform widely recognized for hosting publicly accessible datasets used in data science and academic research. It contains detailed information on users' social media habits, addiction scores, sleep patterns, and mental health indicators. The second dataset was retrieved from GitHub, where researchers and developers commonly publish datasets and analytical resources. This additional dataset includes complementary behavioral attributes, such as platform engagement patterns and digital usage metrics. Together, the two datasets were combined, cleaned, and standardized to form a unified analytical dataset, providing a comprehensive foundation for examining social media usage trends, addiction levels, and their associated health outcomes.

The dataset used in this study consists of survey-based records of social media users across different age groups (Teens, Adults, and Seniors). It includes information such as gender, most-used social media platform, daily usage hours, screen time awareness, sleep duration, mental health score, health effects, and



addiction level. The dataset covers a broad range of respondents, allowing comparative analysis across age groups and usage behaviors.

- **Dataset Coverage**

The dataset includes a wide range of users with different backgrounds and social media habits. While the original data contains responses from several countries, this study focuses only on users from the Philippines. This filtering helps keep the analysis consistent and makes sure the results apply to a group of people with similar cultural and social environments.

The dataset covers important information about the respondents, including their demographics, social media behavior, and health-related indicators. For demographics, it includes users aged 13 to 89 years old, grouped into Teens (13–19), Adults (20–59), and Seniors (60–89). It also lists the respondent's gender and country, though only entries from the Philippines were kept for this study.

In terms of social media behavior, the dataset provides details such as average daily hours spent on social media, the platform they use the most (like Facebook, Instagram, TikTok, YouTube, or Twitter), whether they are aware of their screen time, and the device they usually use (phone, PC, or tablet).

The dataset also includes several health and wellness indicators. These include each user's mental health score, any health effects they experience from social media use, their average sleep hours, and their addiction level, which is given both as a number and as a category (Low, Medium, or High).

Because the dataset contains users of different ages, habits, and health conditions, it provides a strong foundation for studying how social media use affects addiction levels, sleep quality, and overall mental well-being.



- **Type of data (data dictionary)**

DATASET 1

Name	Type	Range	Description
ID	Integer	[1-25000]	Unique identifier for each respondent.
AGE	Integer	[13-49]	Age of the respondent.
GENDER	Nominal	[FEMALE, MALE, female, male]	Self-reported gender of the respondent.
COUNTRY	Nominal	[USA, australia, canada, ph, uk]	Country of residence of the respondent.
AVG DAILY USE HRS	Real	[0.200-12]	Hours spent daily on social media.
MOST USED PLATFORM	Nominal	[BBM, Facebook, Google Plus, MySpace, Skype, Tumblr, Twitter, Yahoo Messenger]	Main social media platform used by the respondent.
SPOTTED ANXIETY	Nominal	[feeling left out, none, overthinking, pressure from peers, self-doubt, social fear]	Reported signs or feelings of anxiety linked to social media use.
REASON FOR USING	Nominal	[chatting, games, join groups, music sharing, posting pics, status updates]	The respondent's main purpose for using social media.



HEALTH EFFECTS	Nominal	[No, Yes]	Whether the respondent reported health-related effects from social media.
AVG SLEEP HRS	Real	[2-12]	Average hours of sleep per day.
MENTAL HEALTH SCORE	Integer	[1-99]	Self-reported mental health score.
ADDICTED LVL	Integer	[1-99]	Self-reported addiction score.
DEVICE USED	Nominal	[BlackBerry, Desktop PC, Laptop, Nokia phone, iPod Touch]	Device typically used to access social media.

DATASET 2

Name	Type	Range	Description
id	Integer	[1-10000]	Unique identifier for each respondent.
age	Integer	[13-59]	Age of the respondent.
gender	Nominal	[Female, Male]	Biological sex of the respondent.
country	Nominal	[AU, CA, PH, UK,US]	Geographic location of the respondent.



avg_daily_use_hrs	Real	[0.500-10]	Hours spent using social media daily.
platform	Nominal	[Facebook, Instagram, TikTok, Twitter, YouTube]	Most frequently used social media platform.
health_effect	Nominal	[Anxiety, Eye strain, Headache, None, Poor posture]	Whether the respondent reported health-related effects from social media.
avg_sleep_hrs	Real	[3-10]	Average hours of sleep per day.
daily_checks	Integer	[1-199]	Times the user clicks an app
avg_session_min	Integer	[1-59]	Average duration (minutes) of each social media session.
primary_reason	Nominal	[Communication, Education, Entertainment, News, Work]	Main reason for using social media.
change_attempts	Integer	[0-9]	Number of times the user tried to reduce usage.
social_ratio	Real	[0-2]	Ratio of social content consumed vs other content.
screen_awareness	Nominal	[No, Yes]	Whether the user is aware of their daily



			screen time.
peak_time	Nominal	[Afternoon, Evening, Late Night, Morning]	Time of day when the user is most active online.
mentalhealth_score	Integer	[1-99]	Self-reported mental health score.
addiction_lvl	Integer	[1-99]	Social media addiction score.
device	Nominal	[Laptop, PC, Phone, Tablet]	Main device used to access social media.

Objectives

The main objectives of this project are to:

1. Compare social media addiction levels based on age group and gender.
2. Evaluate the impact of social media on health and well-being, including average sleep hours, health effects, and mental health scores.
3. Examine the relationship between daily usage habits (average daily use hours and most used platform) and addiction levels.
4. Assess how different social media platforms (e.g., TikTok, Facebook, Instagram) influence addiction severity across age groups.
5. Explore the relationship between screen time awareness and addiction level.

Planned Techniques and Tools Used

This project utilized the following data mining techniques and software tools:



- Decision Tree (Classification): Used to predict social media addiction levels based on demographic and behavioral variables such as age, screen time, and sleep hours.
- Random Forest (Ensemble Classification): Applied to improve prediction accuracy and reduce overfitting by combining multiple decision trees.
- K-Means Clustering (Unsupervised Learning): Used to group users into clusters based on similarities in age, usage behavior, addiction level, sleep hours, and mental health scores.

For visualization and analytical processing, the following tools were used:

- Power BI – for creating dashboards and data visualizations.
- Altair RapidMiner Studio – for data preprocessing, model construction, performance evaluation, and clustering.
- Microsoft Excel / CSV Format – for initial data cleaning and formatting.

These techniques and tools were selected to ensure accurate predictive modeling, meaningful pattern discovery, and effective visual presentation of insights.

II. Data Preprocessing

Overview of Raw Data Issues

The original dataset contained several data quality issues that could affect the reliability of the analysis. These included inconsistent attribute naming, inconsistent text casing, unclear health effect values, unnecessary variables, mixed device categories, and the presence of records from countries outside the study scope. Additionally, raw numeric values for mental health and addiction were not yet grouped into meaningful categories, and the age variable required segmentation into interpretable age groups. Some platform and device values were also inconsistent and required standardization before analysis.

Data Cleaning Steps



Several data cleaning procedures were implemented to ensure accuracy, consistency, and reliability:

1. Standardizing Attribute Names

All selected columns were renamed using a uniform naming format for consistency. The final standardized attributes include: Age, Gender, Country, Average Daily Use Hours, Most Used Platform, Health Effect, Average Sleep Hours, Mental Health Score, Screen Time Awareness, Addiction Level, and Device Used.

2. Fixing Inconsistent Casing

Text values were standardized to ensure uniform capitalization. For example, values such as *female*, *FEMALE* were converted to Female, and *PH*, *philippines* were converted to Philippines.



Fixing Inconsistent Casing

Find and Replace

Find Replace

Find what: FEMALE No Format Set Format...

Replace with: Female No Format Set Format...

Within: Sheet ☐ Match case

Search: By Rows ☐ Match entire cell contents

Look in: Formulas Options <<

Replace All Replace Find All Find Next Close

Find and Replace

Find Replace

Find what: female No Format Set Format...

Replace with: Female No Format Set Format...

Within: Sheet ☐ Match case

Search: By Rows ☐ Match entire cell contents

Look in: Formulas Options <<

Replace All Replace Find All Find Next Close

Find and Replace

Find Replace

Find what: MALE No Format Set Format...

Replace with: Male No Format Set Format...

Within: Sheet ☐ Match case

Search: By Rows ☐ Match entire cell contents

Look in: Formulas Options <<

Replace All Replace Find All Find Next Close

Find and Replace

Find Replace

Find what: australia No Format Set Format...

Replace with: Australia No Format Set Format...

Within: Sheet ☒ Match case

Search: By Rows ☐ Match entire cell contents

Look in: Formulas Options <<

Replace All Replace Find All Find Next Close

Find and Replace

Find Replace

Find what: AU No Format Set Format...

Replace with: Australia No Format Set Format...

Within: Sheet ☒ Match case

Search: By Rows ☐ Match entire cell contents

Look in: Formulas Options <<

Replace All Replace Find All Find Next Close

Find and Replace

Find Replace

Find what: ph No Format Set Format...

Replace with: Philippines No Format Set Format...

Within: Sheet ☒ Match case

Search: By Rows ☐ Match entire cell contents

Look in: Formulas Options <<

Replace All Replace Find All Find Next Close

Find and Replace

Find Replace

Find what: uk No Format Set Format...

Replace with: United Kingdom No Format Set Format...

Within: Sheet ☒ Match case

Search: By Rows ☐ Match entire cell contents

Look in: Formulas Options <<

Replace All Replace Find All Find Next Close

Find and Replace

Find Replace

Find what: US No Format Set Format...

Replace with: United States of America No Format Set Format...

Within: Sheet ☒ Match case

Search: By Rows ☐ Match entire cell contents

Look in: Formulas Options <<

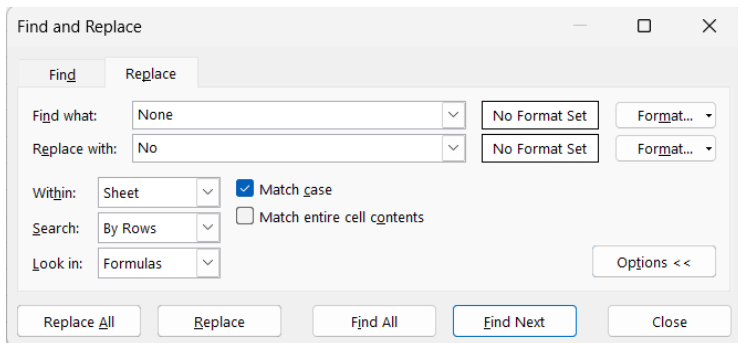
Replace All Replace Find All Find Next Close

3. Handling Health Effect Values

Inconsistent values in the “Health Effect” column were corrected by applying a consistent rule:

- Entries indicating no condition were converted to No
- Entries indicating any issue or symptom were converted to Yes

Handling “Health Effect” Values



Find and Replace

Find Replace

Find what: None No Format Set Format...

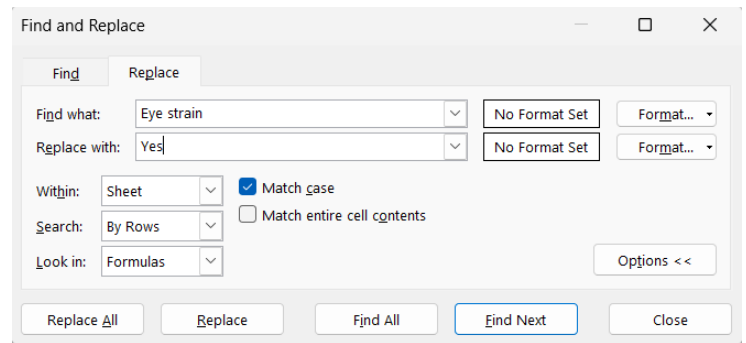
Replace with: No No Format Set Format...

Within: Sheet ☒ Match case ☐ Match entire cell contents

Search: By Rows

Look in: Formulas Options <<

Replace All Replace Find All Find Next Close



Find and Replace

Find Replace

Find what: Eye strain No Format Set Format...

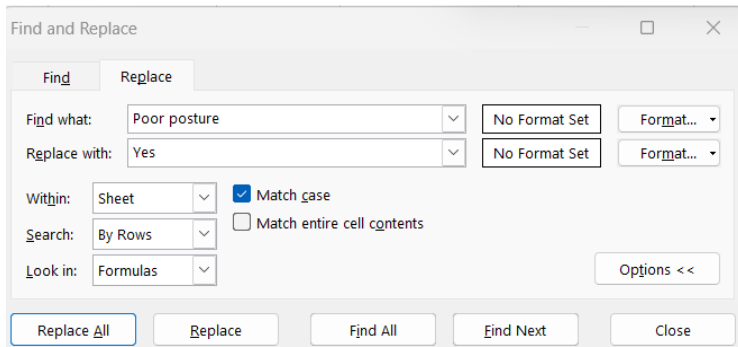
Replace with: Yes No Format Set Format...

Within: Sheet ☒ Match case ☐ Match entire cell contents

Search: By Rows

Look in: Formulas Options <<

Replace All Replace Find All Find Next Close



Find and Replace

Find Replace

Find what: Poor posture No Format Set Format...

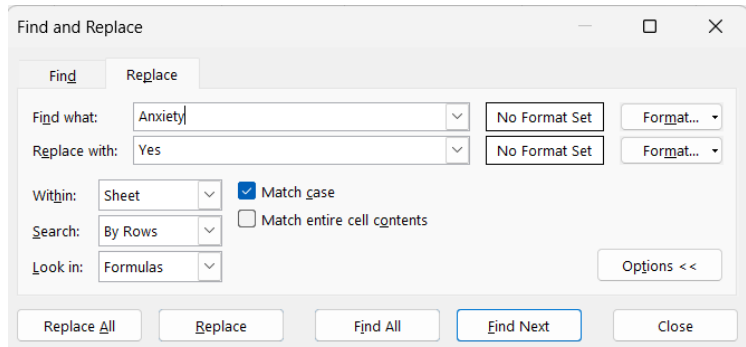
Replace with: Yes No Format Set Format...

Within: Sheet ☒ Match case ☐ Match entire cell contents

Search: By Rows

Look in: Formulas Options <<

Replace All Replace Find All Find Next Close



Find and Replace

Find Replace

Find what: Anxiety No Format Set Format...

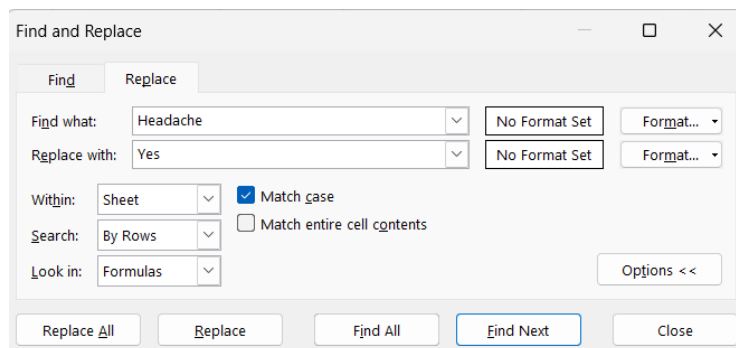
Replace with: Yes No Format Set Format...

Within: Sheet ☒ Match case ☐ Match entire cell contents

Search: By Rows

Look in: Formulas Options <<

Replace All Replace Find All Find Next Close



Find and Replace

Find Replace

Find what: Headache No Format Set Format...

Replace with: Yes No Format Set Format...

Within: Sheet ☒ Match case ☐ Match entire cell contents

Search: By Rows

Look in: Formulas Options <<

Replace All Replace Find All Find Next Close

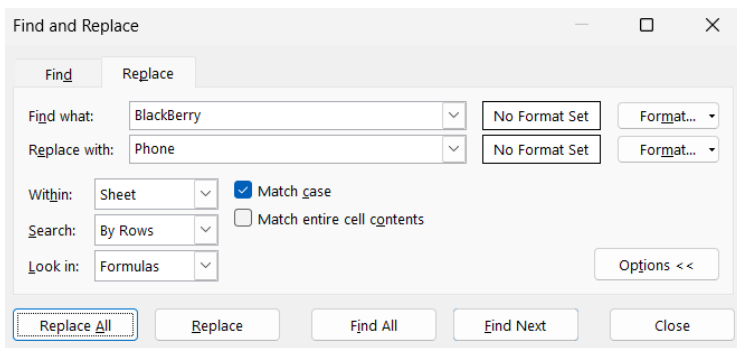
4. Removing Unnecessary Columns

Irrelevant variables that were not part of the study objectives were removed. These included: daily checks, average session minutes, primary reason, change attempt, social ratio, peak time, and spotted anxiety.

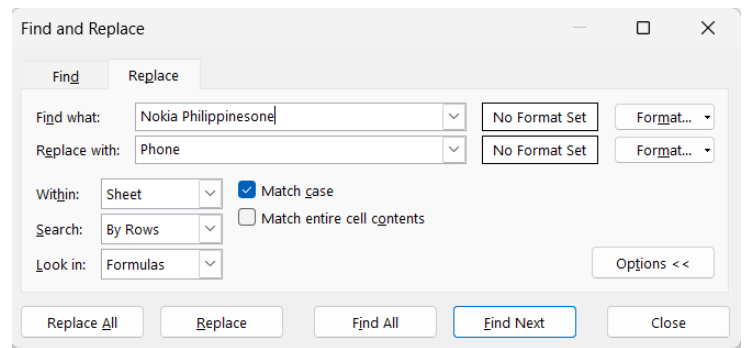
5. Standardizing Device Categories

Device values were consolidated to avoid redundancy. For example, BlackBerry and Nokia were converted to Phone, and Desktop PC was converted to PC.

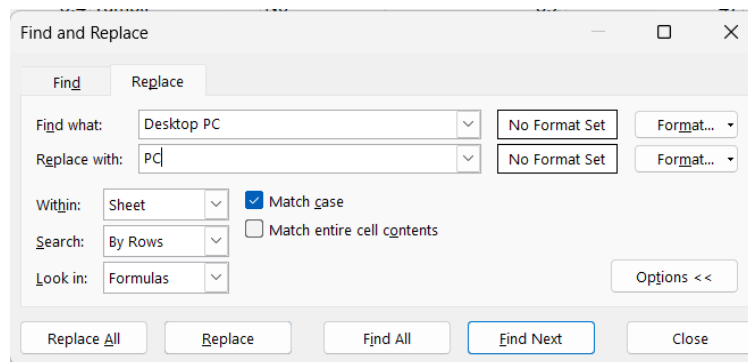
Standardizing Device Categories



Find and Replace dialog box showing the replacement of BlackBerry with Phone. The Find what field contains BlackBerry, and the Replace with field contains Phone. The Match case checkbox is checked. The Search scope is set to By Rows, and the Look in scope is set to Formulas. The Replace All button is highlighted.



Find and Replace dialog box showing the replacement of Nokia Philippinesone with Phone. The Find what field contains Nokia Philippinesone, and the Replace with field contains Phone. The Match case checkbox is checked. The Search scope is set to By Rows, and the Look in scope is set to Formulas. The Find Next button is highlighted.



Find and Replace dialog box showing the replacement of Desktop PC with PC. The Find what field contains Desktop PC, and the Replace with field contains PC. The Match case checkbox is checked. The Search scope is set to By Rows, and the Look in scope is set to Formulas. The Find Next button is highlighted.

6. Filtering Countries

Only records from the Philippines were retained in the dataset. All entries from other countries were removed to maintain geographic consistency.

7. Filtering Most Used Platforms

The dataset was restricted to five platforms only: Facebook, Instagram,



TikTok, YouTube, and Twitter. All records with other platform names were excluded.

Data Transformation

After cleaning, multiple transformation steps were applied to prepare the dataset for analysis:

1. Grouping Mental Health Score and Addiction Level

Raw numeric scores were converted into categorical values for better interpretability:

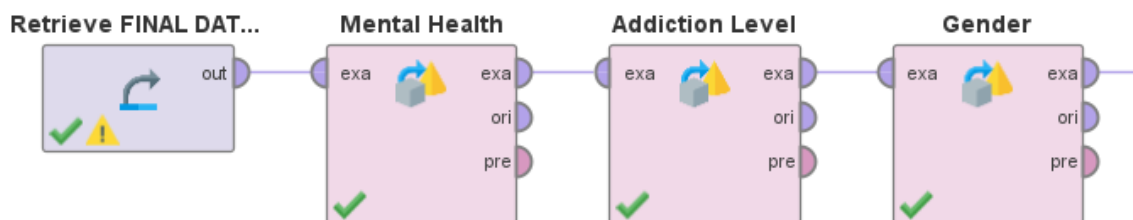
- 1–3 = Low
- 4–6 = Medium
- 7–10 = High

2. Creating Age Groups

The Age variable was transformed into three categorical age groups:

- 13–19 = Teen
- 20–59 = Adult
- 60–89 = Senior

Grouping Mental Health Score & Addiction Level and Creating Age Groups





The first screenshot shows the 'Mental Health (Discretize by User Specification)' parameter. The 'attribute filter type' is set to 'single' and the 'attribute' is 'Mental Health Score'. The 'class names' table has three rows: 'Low' with an 'upper limit' of 3.0, 'Medium' with 6.0, and 'High' with 'Infinity'. The 'Parameters' panel on the right has 'invert selection' and 'include special attributes' unchecked.

class names	upper limit
Low	3.0
Medium	6.0
High	Infinity

The second screenshot shows the 'Addition Level (Discretize by User Specification)' parameter. The 'attribute filter type' is 'single' and the 'attribute' is 'Addition Level'. The 'class names' table has three rows: 'Low' with an 'upper limit' of 33.0, 'Medium' with 66.0, and 'High' with 'Infinity'. The 'Parameters' panel on the right has 'invert selection' and 'include special attributes' unchecked.

class names	upper limit
Low	33.0
Medium	66.0
High	Infinity

The third screenshot shows the 'Discretize (Discretize by User Specification)' parameter. The 'attribute filter type' is 'single' and the 'attribute' is 'Age'. The 'class names' table has three rows: 'Teen' with an 'upper limit' of 19.0, 'Adult' with 59.0, and 'Senior' with 'Infinity'. The 'Parameters' panel on the right has 'invert selection' and 'include special attributes' unchecked.

class names	upper limit
Teen	19.0
Adult	59.0
Senior	Infinity

3. Text Normalization

Extra spaces were trimmed, spelling inconsistencies were corrected, and all categorical values were standardized to ensure uniform formatting across the dataset.

4. Dataset Merging and Structuring

The two datasets were merged into a single unified dataset. Duplicate records were removed, and all attributes were aligned properly. Columns were rearranged into a clean and readable structure for modeling and visualization.

Final Dataset Description

This cleaned dataset represents the final, refined version of the collected survey data, prepared after removing inconsistencies, handling missing values, and standardizing all variable formats. It now provides a reliable foundation for analysis related to digital usage, mental health, and behavioral patterns.



Number of rows after cleaning: 3,379

Number of columns after cleaning: 12 attributes

Final Attributes Used:

- Age (Categorized)
- Gender
- Country
- Average Daily Use Hours
- Most Used Platform
- Health Effect
- Average Sleep Hours
- Mental Health Score (Categorized)
- Screen Time Awareness
- Addiction Level (Categorized)
- Device Used

III. Data Analysis Results

Overview of Techniques Applied

Three major data mining techniques were applied to analyze the social media addiction dataset: Decision Tree, Random Forest, and K-Means Clustering. The Decision Tree is a classification technique that predicts addiction level by creating a set of rule-based conditions derived from input variables such as age, screen time, and sleep hours. It is mainly used for interpretability and understanding how decisions are formed. The Random Forest is an ensemble learning method that builds multiple decision trees and combines their outputs to improve prediction accuracy and reduce overfitting. Lastly, K-Means Clustering is an unsupervised learning technique that groups users into clusters based on similarities in selected variables such as age group, addiction level, daily usage, sleep hours, and mental health scores to identify behavioral patterns across different user segments.

Findings from the Data Mining Techniques Used

1. Decision Tree Results

The Decision Tree model achieved an accuracy of 66.54%, with a classification error of 33.46%, weighted recall of 57.28%, and weighted precision of 64.39%. The confusion matrix showed that the Low addiction class had the highest recall at 79.09%, indicating that the model was very effective at identifying users with low addiction. The Medium addiction class achieved a moderate recall of 57.86%, while the High addiction class had a low recall of only 34.88%, meaning that many highly addicted users were misclassified as Medium or Low. Overall, the Decision Tree performed moderately well but struggled particularly with accurately predicting the High addiction group due to overlapping feature patterns and class imbalance.

2. Random Forest Results

The Random Forest model outperformed the Decision Tree with an accuracy of 68.90%, a classification error of 31.10%, weighted recall of 59.01%, and weighted precision of 66.71%. The confusion matrix showed that the Low addiction level was predicted very well with an 86.00% recall. The Medium addiction group achieved a 55.00% recall, while the High addiction class slightly improved to 36.05% recall compared to the Decision Tree. These results show that Random Forest provided more stable and reliable predictions by reducing overfitting and handling noisy data better than a single tree model.

3. K-Means Clustering Results

The K-Means algorithm grouped the dataset into three distinct clusters:

- Cluster 0 – Moderate Users with Balanced Lifestyle: This group consisted mostly of Adults (0.672) and some Seniors (0.264), with a dominant Medium addiction level (0.563) and High mental health score (0.621). The average daily usage was 4.98 hours, and the average sleep duration was 7.33 hours, indicating controlled and balanced digital behavior.
- Cluster 1 – High Usage and High Addiction Group: This cluster contained a mix of Adults (0.662) and Teens (0.330). It showed the highest average



daily use at 8.11 hours, the lowest sleep duration at 5.95 hours, and the highest proportion of High addiction levels. This group is classified as the most at-risk cluster.

- Cluster 2 – Low Usage and Healthy Lifestyle Group: Dominated by Seniors (0.674), this group showed a very high dominance of Low addiction level (0.797), high mental health score (0.614), lowest daily use at 2.74 hours, and highest average sleep at 8.82 hours, indicating very healthy digital behavior.

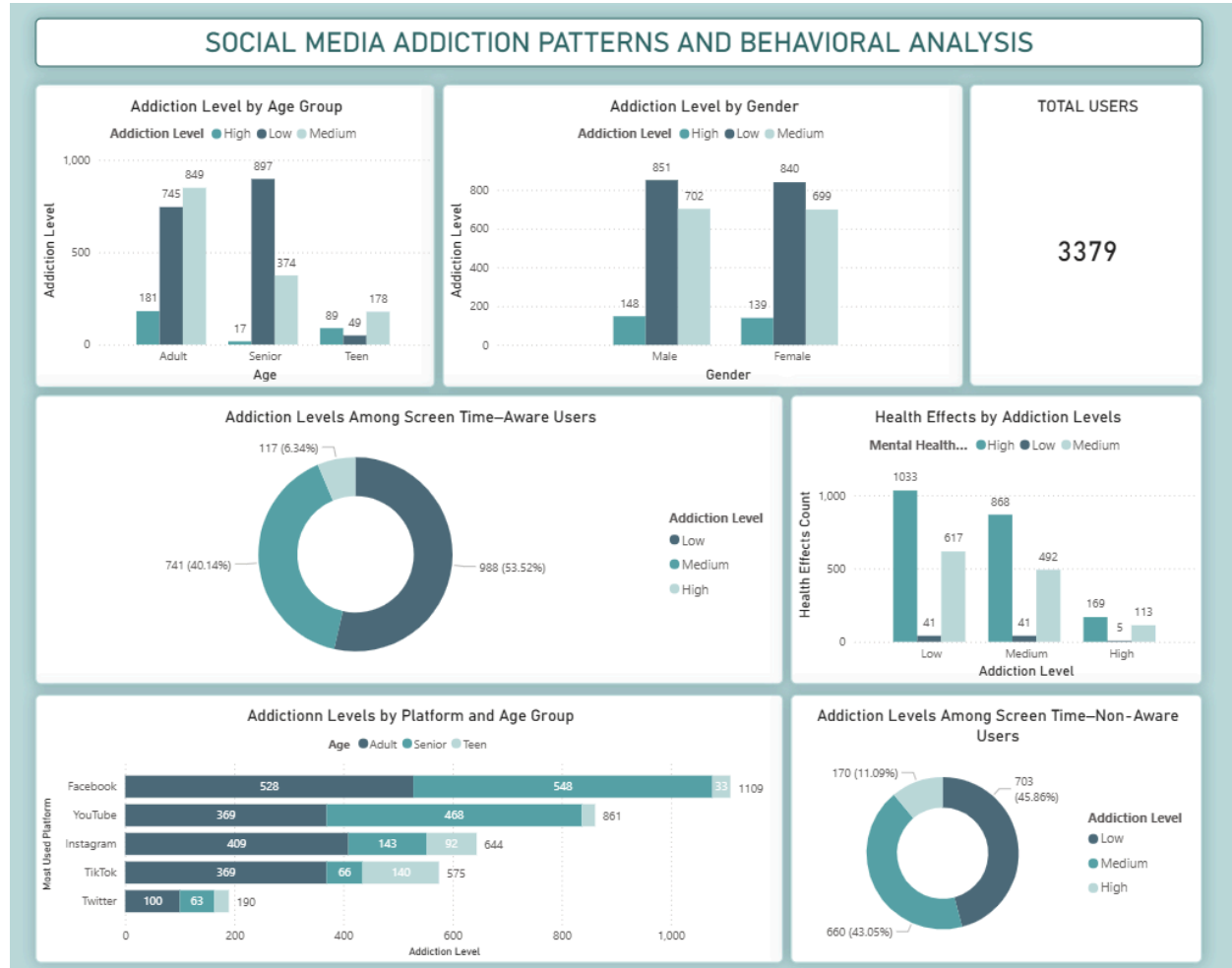
Overall clustering showed that higher screen time is linked to higher addiction and reduced sleep, while users with low screen time displayed healthier mental and sleep patterns.

Summary of Key Insights

The results clearly indicate that Random Forest is the strongest predictive model, achieving the highest accuracy and most stable performance among the classification techniques. Both classification and clustering results consistently show that long daily screen time and shorter sleep duration significantly increase the likelihood of high social media addiction. The clustering analysis identified a high-risk user group (Cluster 1) composed mainly of teens and adults with excessive usage and poor sleep habits. On the other hand, seniors with low screen exposure consistently exhibit better mental health and lower addiction levels. These findings can be used to support early intervention programs, digital wellness campaigns, and policy development focused especially on younger high-risk users.

IV. Visualizations

Visualization 1 - Dashboard



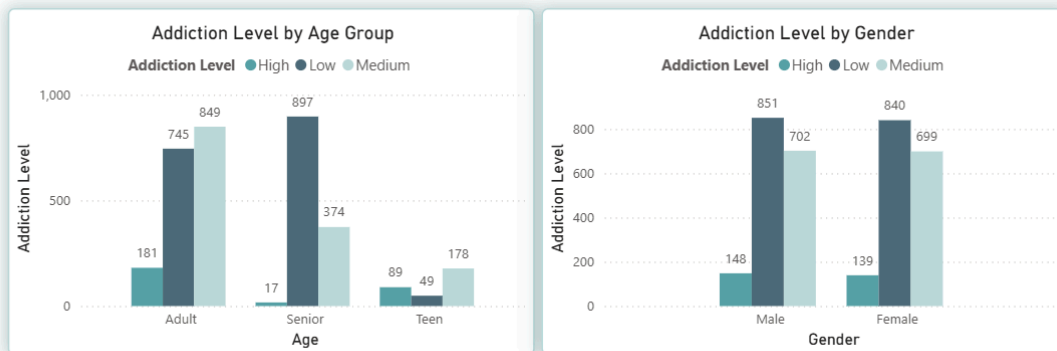
The dashboard provides a comprehensive visual summary of the key variables affecting social media addiction, including age group, gender, platform usage, screen time awareness, mental health scores, sleep hours, and addiction levels. It allows users to quickly observe overall trends and patterns across the entire dataset. The dashboard highlights that low addiction levels dominate across most demographic groups, particularly among seniors, while adults show a higher concentration of medium addiction, and teens exhibit relatively more high-addiction cases compared to other age groups. Gender comparisons reveal that both males and females follow nearly identical addiction patterns, with low addiction being the most common for both. This visualization effectively serves as a high-level decision-support tool for identifying risk groups and behavioral trends at a glance.



Visualization 2 - Detailed Analytical Visualization

This section presents detailed analytical charts that directly answer the key research objectives of the study, specifically the relationships between addiction level, mental health, sleep duration, social media platform usage, and age group. Each visualization highlights how behavioral factors influence social media addiction severity.

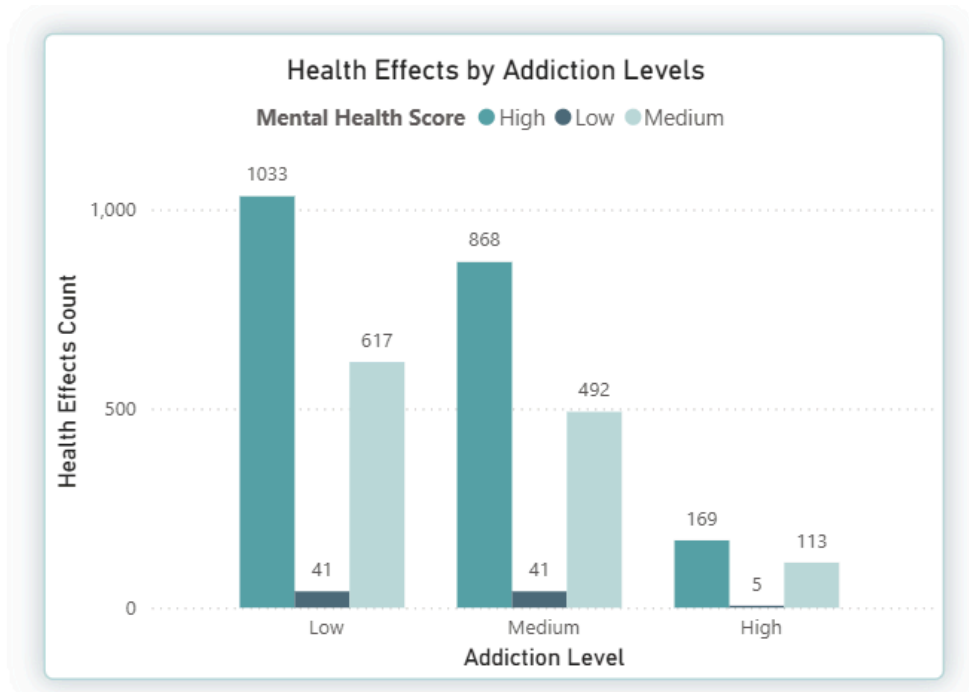
- **Compare social media addiction levels based on age group and gender.**



The data shows that low addiction levels are the most common across both age groups and genders. Seniors have the highest number of low-addiction cases, while adults tend to have more medium-level addiction. Teens show fewer cases overall but have a relatively higher number of high-addiction cases compared to other ages. For gender, males and females follow almost the same pattern: low addiction is the most common, medium comes next, and high addiction is the least frequent for both.



- *Evaluate the impact of social media on health and well-being, including average sleep hours, health effects, and mental health scores.*



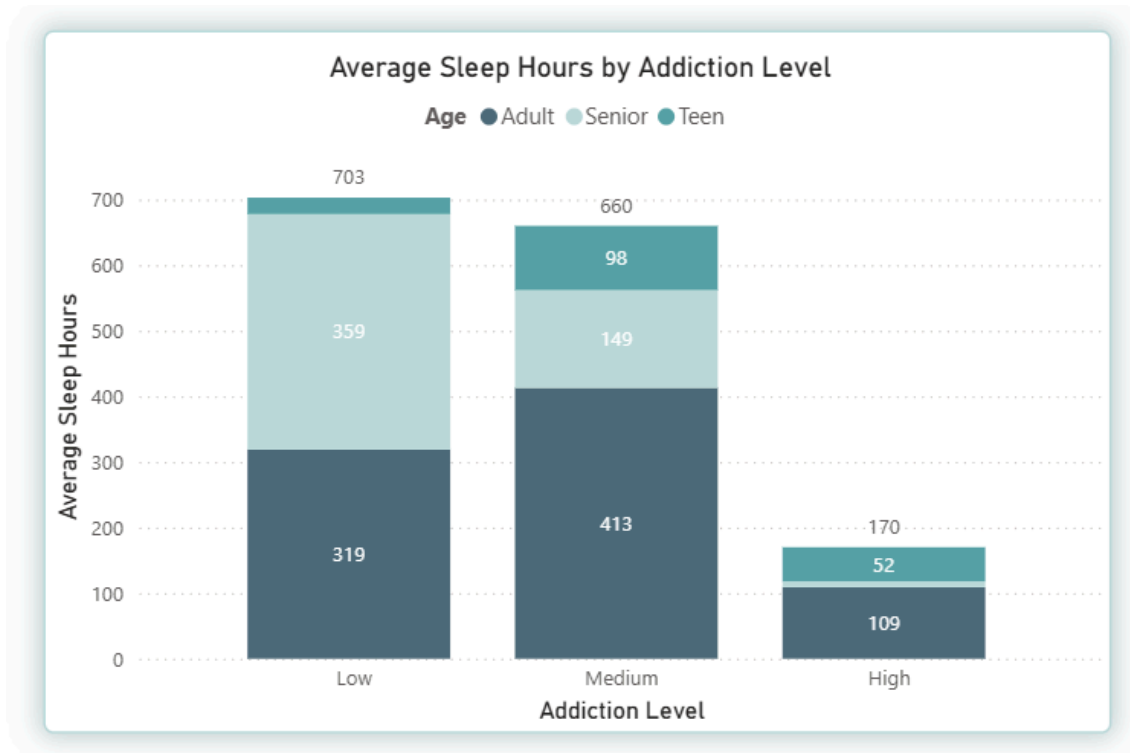
The grouped bar chart shows a clear trend: higher social media addiction is linked to poorer mental health.

- **Low Addiction Group:** Most users have Medium (868) or High (617) mental health scores, while fewer report Low scores (1,003).
- **High Addiction Group:** The majority report Low mental health scores (192), with fewer in Medium (41) or High (169) categories.

Higher social media addiction is strongly associated with worse self-reported mental health. The chart shows that as addiction increases, negative health effects (Low mental health scores) become more prevalent.



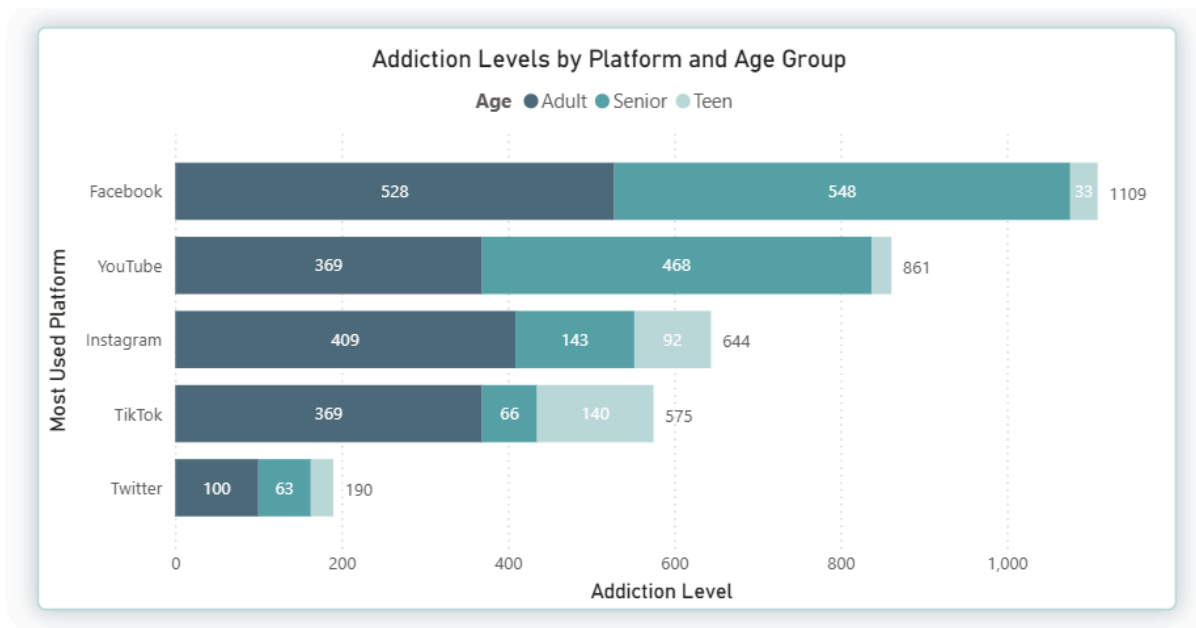
- **Examine the relationship between daily usage habits (average daily use hours and most used platform) and addiction levels.**



This data shows a clear relationship between addiction level and average sleep hours across different age groups. Overall, higher addiction levels are linked to significantly less sleep. This pattern is strongest among teenagers: teens with low addiction sleep a very high average of 703 hours (which may represent annual or total hours in a period), but those with high addiction sleep only 52 hours on average. Adults and seniors also sleep less as addiction increases, though less dramatically. For example, adults drop from 600 hours (low addiction) to 109 hours (high addiction). In simple terms, the data suggests that higher addiction severely reduces sleep time, with teenagers being the most affected.

Visualization 3 - Insight-Focused Visualization

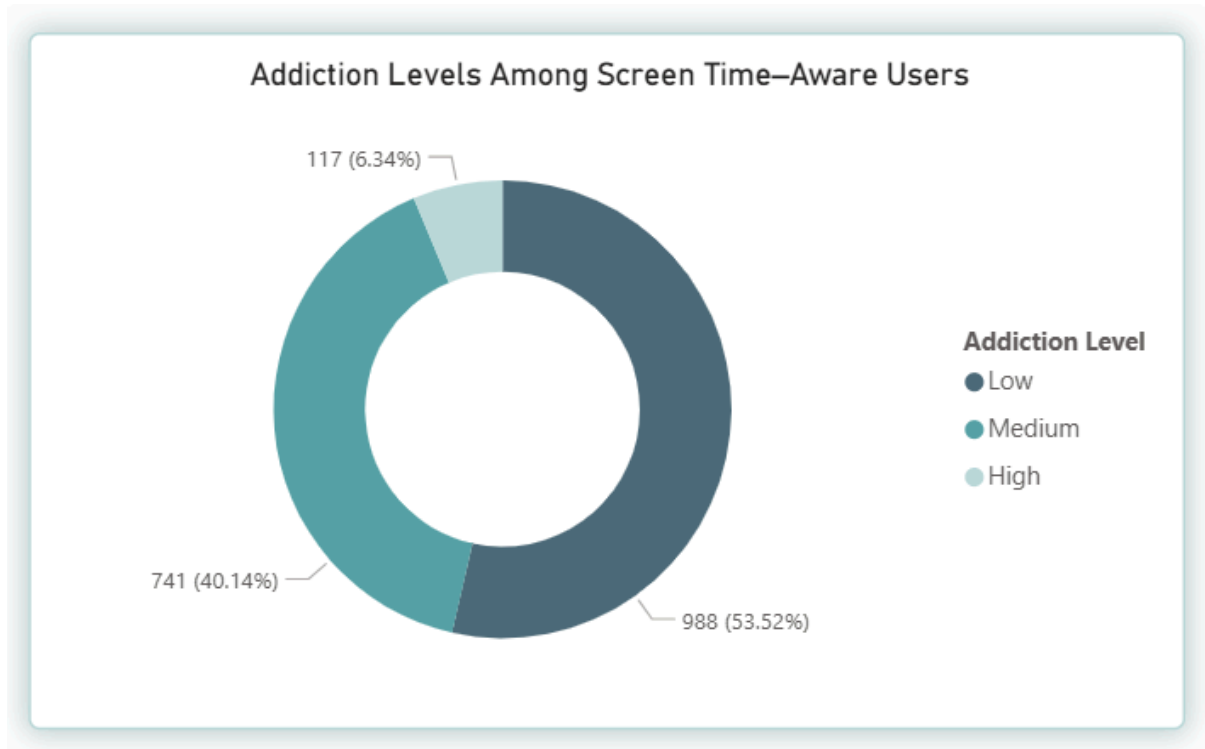
- Assess how different social media platforms (e.g., TikTok, Facebook, Instagram) influence addiction severity across age groups.



The chart illustrates that the association between social media platforms and levels of problematic use is moderated by age. Among teenagers, visual and short-form video platforms, such as TikTok and Instagram, are linked to the highest levels of addictive engagement. In contrast, adults primarily use platforms like YouTube and Facebook, which are associated with more moderate levels of addictive behavior. For seniors, Facebook is the dominant platform, yet it corresponds to relatively low levels of addiction. These findings indicate that the severity of social media addiction is shaped by the interaction between platform characteristics and user age, emphasizing the need for age-sensitive approaches in both research and intervention strategies.



Explore the relationship between screen time awareness and addiction level.



This data shows that among people who are aware of their screen time, addiction levels are quite high. The vast majority, over half (53.52%), fall into the "High" addiction category. A significant portion, 40.14%, have a "Medium" level of addiction. Only a very small group, about 6.34%, are considered to have a "Low" level of addiction. In simple terms, for users who monitor their screen use, high addiction is the most common outcome by a large margin.

V. Conclusion and Recommendations

Summary of Findings

This study applied multiple data mining and visualization techniques to analyze patterns of social media addiction across different demographic and behavioral factors. The classification models showed that the Random Forest algorithm achieved the highest predictive accuracy (68.90%), outperforming the Decision Tree model. Both models revealed that screen time, sleep hours, and age are strong predictors of



addiction level. The clustering analysis further identified three distinct user groups: moderate users with balanced lifestyles, high-usage users with severe addiction risk, and low-usage users with healthy behavioral patterns.

Visual analysis confirmed that higher addiction levels are strongly associated with poorer mental health and reduced sleep duration, with teenagers being the most affected group. Platform-based analysis revealed that TikTok and Instagram are most strongly linked to high addiction levels, particularly among younger users, while seniors using Facebook show generally low addiction levels. Additionally, results showed that screen time awareness alone does not significantly reduce addiction, as a majority of screen-time-aware users still fall under medium to high addiction categories. Overall, the findings consistently demonstrate that excessive social media use negatively impacts both mental health and sleep quality, especially among younger populations.

Recommendations

Based on the findings of this study, the following actionable recommendations are proposed:

1. Implement Digital Wellness Programs for Students and Teens

Schools and local institutions should introduce structured digital well-being education that promotes responsible screen use, healthy sleep habits, and self-regulation strategies, especially targeting teenagers who are at the highest risk of addiction.

2. Encourage Platform-Based Usage Controls

Social media platforms should strengthen features such as screen time limits, automatic usage alerts, and content moderation, particularly for highly engaging platforms like TikTok and Instagram.

3. Promote Parental and Family Involvement

Parents should actively monitor and guide younger users' online behavior by setting time boundaries, encouraging offline activities, and fostering awareness of the mental health risks of excessive social media use.



4. Support Mental Health Interventions

Healthcare providers and guidance counselors should integrate early screening for social media addiction into routine mental health assessments, especially for students exhibiting sleep problems or emotional distress.

Limitations of the Study

Although the study produced valuable insights into social media addiction, several limitations should be considered when interpreting the results. First, the dataset is based on self-reported information, which may contain inaccuracies or subjective responses. Variables such as average daily use hours, sleep duration, mental health score, and addiction level depend on personal perception, making them prone to reporting bias.

Second, the dataset does not include any longitudinal or time-based measurements. Since all records represent a single point in time, the study cannot track behavioral changes or determine how social media usage patterns develop over longer periods. This limits the ability to establish causal relationships between digital habits and well-being.

Third, while the dataset covers essential demographic and behavioral attributes, several important factors were not captured. Variables such as socioeconomic status, academic performance, family environment, psychological history, and clinical mental health assessments were not included but could significantly influence both screen time behavior and addiction severity. Their absence restricts the depth of the analysis.

Additionally, several preprocessing steps—such as filtering respondents to those from the Philippines, removing uncommon platforms, consolidating device categories, and converting scores into categorical levels—reduced some of the dataset's variability. Although these steps were necessary for consistency, they may have simplified certain patterns that could have been more meaningful in their original form.

Another significant limitation is the presence of class imbalance, particularly in the High Addiction category. This imbalance affected the performance of the classification models, as shown by the lower recall rates for high-addiction predictions in



both the Decision Tree and Random Forest models. Because fewer high-addiction cases were available, the models had difficulty learning their characteristics accurately.

Lastly, the dataset does not include contextual factors such as internet quality, access to digital devices, or environmental influences that may affect usage habits. These missing elements could help explain variations in screen time and addiction levels but were outside the scope of the available data.

Overall, these limitations indicate that future research would benefit from more diverse variables, balanced category distributions, larger sample sizes, and the inclusion of real-time behavioral data. Incorporating these elements would strengthen the accuracy, reliability, and generalizability of the study's findings.

VI. Appendices

Power BI File Link

[BIDMFINALS.pbix](#)

Raw and Cleaned Data Set Samples

RAW:

[DATASET1-SOCMED-ADDICTION.csv](#)

[DATASET2-SOCMED-ADDICTION.csv](#)

CLEAN:

[BIDM-FINALDATASET.csv](#)

Self Reflection for the Project

Escaño, Krisha Ann Ame:

This project showed me how important data cleaning, preparation, and validation are before doing any real analysis. I realized that even simple tasks like fixing inconsistent values, grouping variables, and merging datasets can have a big effect on the accuracy of the results. Using Decision Tree, Random Forest, and K Means taught me that different models can provide different insights and that no single method works best for every situation. The visualizations also helped me understand patterns in social media use, especially how screen time, age, sleep, and mental health connect to



addiction levels. Overall, this project improved my skills in analyzing data and made me more aware of how social media habits can affect people in different ways.

Suico, Agnes Catherine:

Working on this project allowed me to gain hands-on experience in applying data mining techniques such as Decision Tree, Random Forest, and K-Means Clustering to a real dataset. I learned how to analyze model performance, interpret clustering results, and communicate insights through effective visualizations. The project also taught me the value of collaboration, as each member contributed unique strengths to achieve a common goal. Despite facing challenges related to data understanding and visualization design, I was able to improve my critical thinking and technical skills. This experience has prepared me to approach future analytical projects with greater confidence and responsibility.